

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
1	(a)	(i)		- Corpus luteum (1) - Progesterone secretion (1) Ignore oestrogen Reject FSH/ LH	2			2		2
		(ii)		provides materials for hormone {synthesis/secretion/ production} /so the hormones can be transported (1)		1		1		
	(b)	(i)		Award 2 marks for 0.0368 to 0.0380 Accept correct rounding 3.68×10^{-2} to 3.80×10^{-2} If incorrect award 1 mark for $150/4 \times 1/1000$ Allow 147 to 150		2		2	2	
		(ii)		426 to 435 times larger Ecf from I 16/ answer from (i)		1		1	1	

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	(c)	(i)		1 mark per row <table border="1"><thead><tr><th>Letter</th><th>Name of structure</th><th>Function during fertilisation</th></tr></thead><tbody><tr><td>X</td><td>Acrosome</td><td>Secretes an enzyme to digest the {zona pellucida/jelly coat/outer layer of (2°) oocyte} <i>not corona radiata alone</i></td></tr><tr><td>Y</td><td>Nucleus/ head</td><td>carries {genetic material/ chromosomes/DNA}/ fuses with {female nucleus/ (2°) oocyte} to form {2n/zygote}</td></tr><tr><td>Z</td><td>Mito- chondrion/ia</td><td>provides <u>ATP</u> for {motility/movement} Ignore energy</td></tr><tr><td>OR Z</td><td>Middle piece/ part</td><td>Contains mitochondria which provides <u>ATP</u> for {motility/movement}</td></tr></tbody></table> If no marks awarded for correct rows award 1 mark for three correct structures.	Letter	Name of structure	Function during fertilisation	X	Acrosome	Secretes an enzyme to digest the {zona pellucida/jelly coat/outer layer of (2°) oocyte} <i>not corona radiata alone</i>	Y	Nucleus/ head	carries {genetic material/ chromosomes/DNA}/ fuses with {female nucleus/ (2°) oocyte} to form {2n/zygote}	Z	Mito- chondrion/ia	provides <u>ATP</u> for {motility/movement} Ignore energy	OR Z	Middle piece/ part	Contains mitochondria which provides <u>ATP</u> for {motility/movement}							
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		(ii)		69 (1) - Triploid/ Triploidy (1) Reject polyspermy/ polyploidy/ trisomy		1	1	2																		
		(iii)		- Endosperm (1) - as {food /starch/protein/lipid} (source) (1) ignore nutrients	2			2																		

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	(d)			Any two (×1) from: • has a large surface area + {diffusion/absorption/ gas exchange/ eq} (1) • {Good blood supply/many arteries/ many veins} and {to carry nutrients/waste} (1) • Umbilical cord + carries blood {to/ from} the {placenta/ embryo/ foetus/ baby}(1) • Acts as a barrier + to {mixing blood/different blood pressures/pathogen filter/ mother's immune system} or secretes hormones + to maintain pregnancy (1)	2			2		
				Question 1 total	6	8	1	15	3	2